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To the Editors,
Transportmetrica B: Transport Dynamics

Dear Editors,

I am submitting the manuscript “Fair and Transferable Graph Reinforcement Learning for Metro Network Expansion in Asian Cities” for consideration as a regular research article in *Transportmetrica B: Transport Dynamics*. This is a single-anonymized submission per the journal’s policy; I understand the manuscript will be assessed for suitability by the editor and then reviewed by one independent, anonymous expert.

The paper addresses the Metro Network Expansion Problem (MNEP), the rail-specific case of the Transport Network Design Problem (TNDP): given an existing rail network and a limited construction budget, decide which stations or line segments to add next. Classical TNDP solution methods – mixed-integer and bilevel programming, genetic algorithms, tabu search, ant colony optimization – buy tractability by fixing candidate corridors or line counts by hand before optimization begins. Reformulating MNEP as a sequential decision process solvable by reinforcement learning removes that pre-constraint; this journal has itself recently published closely related work applying deep RL directly to transit network design and improvement (Holliday, El-Geneidy, and Dudek, 2025), which we take as a strong signal of fit for a methods contribution of this kind, and three further recent papers have each taken a piece of that reformulation and shown it works: a GNN-based spatial state representation (MetroGNN), a fairness-aware multi-objective reward decomposed via Tchebycheff scalarization, and a tabular reinforcement-learning method that argues deep RL’s overhead is unnecessary at this problem’s scale. No method combines the first two, and none has been tested from small, data-sparse emerging metro systems to the largest networks in the world. We propose a GNN-based, multi-objective actor-critic policy that combines a heterogeneous spatial-graph encoder with a Tchebycheff-scalarized reward over demand, equity, and accessibility, condition the policy itself on the sampled objective-weight vector so one trained model spans the full efficiency-equity trade-off surface, and add a cross-city curriculum-transfer mechanism – pretraining on one city’s network and warm-starting another’s – that responds directly to the training-cost critique of GNN-based methods without abandoning the GNN.

The paper’s contribution is threefold: (1) the first MNEP method to combine GNN spatial representation with multi-objective fairness decomposition in a single policy; (2) a cross-city curriculum and sample-efficiency design (prioritized replay, reward shaping) that directly answers the training-cost and carbon-footprint critique raised against GNN-based MNEP methods; and (3) a rail-only scope discipline – using each city’s existing bus network purely as an auxiliary input signal, never as an action – that lets the method extend, for the first time in this literature, to small, data-sparse emerging metro systems (Hanoi, Ho Chi Minh City) that prior MNEP studies have not evaluated on.

The empirical results are reported as found rather than smoothed toward the hoped-for conclusion. A five-seed sweep on Xi’an finds no reliable advantage for the full method over either architectural ablation on demand, equity, or coverage, and the tabular RL baseline still wins on raw demand under a matched training budget – a negative result for model complexity that we report plainly rather than obscure. Tracing an earlier, more favorable-looking equity finding, we discovered GPU training non-determinism severe enough to invert which architecture a metric appears to favor,

traced it to two distinct causes (non-deterministic scatter reduction in the GNN encoder and an unseeded environment reset), fixed both, and verified deterministic re-runs – a reproducibility hazard we believe generalizes beyond our own implementation to other GNN-based methods built on the same default scatter-reduction operations, and one we think is of independent interest to this journal’s readership regardless of this paper’s own architectural comparisons. On Ho Chi Minh City’s structurally distinct, irregular polygon graph, an encoder ablation collapses to near-zero demand where the full method does not, and warm-starting from Xi’an’s trained policy shows genuine positive transfer at initialization and, under a reduced training budget, reaches non-trivial demand and consistently higher coverage than training from scratch – direct evidence for the curriculum’s sample-efficiency claim on the one city where we could test it. A separately implemented radiation-model accessibility measure does not share the coverage proxy’s architectural sensitivity, a construct-validity finding we report rather than omit.

All code – the GNN encoder, multi-objective policy, training and evaluation scripts for both cities, and the full test suite – is publicly available now, not embargoed until publication, at https://github.com/vinhqdang/metro_expansion. Data sources are public third-party releases (Michailidis et al.’s `mo-tndp`, the CityScope/MIT CSL_HCMC repository, and GZUPA/subway-traffic-data-set), cited and documented in the manuscript’s Data Availability statement.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. I am the sole author and approve its submission. I declare no competing financial or personal interests, and the work received no specific grant from any funding agency.

The main text runs to approximately 14,200 words (Introduction through Conclusion, excluding tables, figures, and references), above the journal’s typical 10,000-word guideline; this reflects two fully reported case-study cities together with the reproducibility and construct-validity findings described above. I would welcome the editor’s guidance on any length reduction expected before the manuscript proceeds to review.

Thank you for your consideration.

Yours sincerely,

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